

Blockchain in Practice

Two Video Overviews: Polkadot and Kubernetes

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2026

Blockchain Course

- Two videos from the course material are summarised: content and key concepts only (no rating).
- **Video 1**, from the course playlist: *What is Polkadot? A Polkadot for Beginners Guide and Intro to Blockchain* (entry 24).
- **Video 2**, free choice: *Managing and Securing Blockchain Applications on Kubernetes* (Haining Zhang & Yang Yu, VMware; KubeCon + CloudNativeCon China 2018).
- Required viewing only, not reviewed: *How the Blockchain is Changing Money and Business*.

Each video is presented as a short overview plus its main concepts and 2 to 3 keywords.

Video 1: What It Is About

Talk at a glance

- Channel: *Polkadot*; published May 2020; runtime $\approx$ 76 min.
- A beginner, jargon-free introduction to Polkadot and blockchain.
- Presenters:
 - **Bill Laboon**: Technical Education Lead, Web3 Foundation.
 - **Dan Reecer**: Community & Growth, Web3 Foundation.

Roadmap stated up front

1. How the Internet came about and evolved.
2. History of blockchain and crypto, and the similarities.
3. What blockchain is.
4. Problems of legacy blockchains.
5. How Polkadot solves them.
6. What to expect from Polkadot long term.

Video 1: From the Internet to the Blockchain

Internet history (the parallel)

- Late 1960s: computers were “islands unto themselves”: data moved by diskette, tape, punch cards.
- **ARPANET**: first network of computing sites (universities CMU, Rutgers, Illinois, plus e.g. NSA).
- Early file-sharing and email; Queen Elizabeth II sent her first email in 1976.
- **1978**: Robert Kahn’s TCP/IP let separate networks share one language, giving birth to the Internet.

What a block is

- “Chunks of data ...associated with a particular time period ...connected in a chain and distributed among many computers.”
- No central place where the blockchain lives; it cannot be altered once the computers agree on the next block.
- Analogies: every poker hand recorded by all players; a chess move-sheet anyone can replay.

Chain of history: 1980s digital cash (eCash) → 1997 Hashcash (Adam Back) → 2008 Bitcoin paper / 2009 network → 2015 Ethereum (programmable apps).

Video 1: Key Concepts & Keywords

Decentralized system

- No central point of storage; many computers maintain one shared chain.
- Lets people trust each other without knowing each other; very hard to shut down.

Problems of legacy chains

- Poor interoperability: good within one chain, weak across networks.
- One size fits all (no customisation); weak governance; hard upgrades.
- Limited scalability: Bitcoin $\approx$ 3 to 4 TPS vs. Visa $>$ 1,700; high fees when busy.

How Polkadot solves them

- **Relay chain + parachains**: many custom chains (IoT, finance, gaming) into one “organism”.
- **Shared security**: validators shared across chains; collators feed data to them.
- Runtime upgrades by on-chain governance: proposal $\rightarrow$ second $\rightarrow$ referendum $\rightarrow$ majority of stake.
- *Substrate* framework to build chains; *Kusama* as a canary network.

Keywords: Internet · Definition of Block · Decentralized System.

Video 2: What It Is About

- **Talk:** *Managing and Securing Blockchain Applications on Kubernetes.*
- **Venue:** KubeCon + CloudNativeCon China 2018 (Nov 14, 2018).
- **Speakers (VMware China R&D):**
 - **Haining “Henry” Zhang:** Technical Director; founder of the Harbor project.
 - **Yang Yu:** Software Engineer; Kubernetes CNI plugin / NSX; ex-OpenStack.
- **Premise:** because blockchain is *decentralised*, the cloud is an ideal place to run it, and Kubernetes is a *ubiquitous cloud operating system*.
- **Goal:** reduce operational complexity by automating enterprise blockchain deployment.

Video 2: Key Concepts & Keywords

Enterprise / private blockchain

- **Hyperledger Fabric**: an open-source, permissioned blockchain framework.
- Aimed at business problems, not public cryptocurrency use.

Automated deployment on Kubernetes

- Deploy Hyperledger Fabric automatically on Kubernetes.
- Kubernetes as the ubiquitous cloud OS reduces operational complexity.

Security by isolation

- Use a **CNI plugin** to apply network policies based on Kubernetes *namespaces*.
- Isolates workloads between blockchain entities (peers, orderers, organisations).

Outcome

- A foundation for enterprise **Blockchain-as-a-Service (BaaS)** on Kubernetes.

Keywords: **Blockchain in Kubernetes** · **Enterprise Scale Blockchain** · **Private Blockchain**.

Video 2: Real-World Adopters & Applications

Walmart & Sam's Club: food traceability at enterprise scale

- Sept 2018: leafy-greens suppliers must give real-time, end-to-end traceability from farm to store on a shared blockchain (built with IBM and 11 food companies; IBM Food Trust).
- Suppliers log digital traceability events; they need only a smart device and internet.
- Earlier pilots (2016) traced Chinese pork and fresh mangoes back to source.

Practical impact

- Store-to-farm tracing: from ~7 days to 2.2 seconds.
- Faster, precise recalls: isolate the affected lot, protect customers, cut waste and losses.
- Same pattern (a shared, permissioned ledger at enterprise scale) is this video's theme.

Who else uses Hyperledger: IBM Food Trust (Nestlé, Unilever); Maersk/IBM TradeLens (MSC, CMA CGM; retired 2022); Bank of Canada bond pilot; Besu: DTCC, Citi, Standard Chartered.

Keywords Recap

Video	Keywords
What is Polkadot? (Beginners Guide)	Internet; Definition of Block; Decentralized System
Managing & Securing Blockchain on Kubernetes	Blockchain in Kubernetes; Enterprise-Scale Blockchain; Private Blockchain

- Video 1, conceptual foundation: what a block is and why decentralisation matters.
- Video 2, putting it into production: permissioned blockchain, automated on Kubernetes, secured by isolation.

